Analyzing Netflix's Content Strategy: A Quantitative Study of Catalog Composition and Predictive Classification

Kayauna Allen

Colorado State University Global

MIS581 - Capstone - Business Intelligence and Data Analytics

Jamia Mills

February 28th, 2026

Abstract

The rapid growth of subscription-based streaming platforms has transformed the global entertainment industry, positioning content strategy as a critical driver of competitive advantage. This study examines Netflix's content strategy through quantitative analysis of publicly available title-level metadata, addressing the challenge of limited access to proprietary performance metrics such as viewership and revenue. Using secondary data and a business intelligence framework, the research analyzes trends in content type distribution over time, evaluates genre frequency within Netflix's catalog, and assesses whether structured metadata attributes can predict content classification as a movie or television show. Descriptive statistics, inferential testing, and predictive modeling were conducted using SAS Visual Analytics. Findings indicate that while both movies and television shows increased substantially over time, movies consistently outnumbered television shows; therefore, the null hypothesis was not rejected for content type dominance. Genre analysis revealed a highly uneven distribution, with comedy, drama, and documentary content appearing significantly more frequently than other genres, therefore, the null hypothesis of genre uniformity was rejected. Predictive modeling demonstrated that metadata attributes such as genre, runtime, and release year provide meaningful predictive power for content classification, leading to rejection of the corresponding null hypothesis. Overall, the study demonstrates that publicly accessible metadata can yield valuable strategic insights and highlights the practical application of business intelligence techniques for evaluating digital platform strategy in data-constrained environments.

Introduction

Over the past decade, the global entertainment industry has experienced a significant transformation driven by the rapid growth of subscription-based streaming platforms. Traditional cable television and physical media formats have steadily declined as consumers increasingly prefer on-demand, internet-based services that offer flexibility, personalization, and global accessibility. Among these platforms, Netflix has emerged as a dominant industry leader, serving hundreds of millions of subscribers across numerous countries. Netflix's sustained success has been closely linked to its ability to leverage data analytics to guide content acquisition, original programming decisions, and recommendation strategies that enhance user engagement and retention.

Netflix's competitive advantage is fundamentally rooted in its content strategy. Decisions regarding the balance between movies and television shows, genre diversity, release timing, and catalog expansion directly influence subscriber growth and long-term platform viability. As competition intensifies with other major streaming services such as Amazon Prime Video, Hulu, Disney Plus, and HBO Max, content strategy has become a critical differentiator in the increasingly crowded streaming marketplace. Strategic misalignment in content offerings can result in subscriber churn, while data-informed content planning can strengthen market positioning and sustain competitive advantage.

Although Netflix is widely recognized for its sophisticated use of analytics, much of its internal performance data, including viewership metrics, completion rates, and revenue per title, remains proprietary. This lack of transparency presents challenges for external researchers attempting to evaluate Netflix's strategic decisions using traditional performance indicators.

However, Netflix does make detailed title-level metadata publicly available, providing an alternative opportunity for empirical analysis. This metadata includes information such as content type, release year, duration, rating, genre classification, and country of production.

As competition intensifies, small shifts in content composition may have outsized effects on customer retention and brand positioning. Unlike traditional broadcasters constrained by linear scheduling and geographic boundaries, streaming platforms compete through continuously evolving digital catalogs that must satisfy diverse audience preferences across global markets. In this environment, content strategy functions as a form of portfolio management, requiring firms to balance scale, variety, and consistency while managing rising production costs and subscriber expectations.

Content strategy within streaming platforms also reflects broader operational constraints and strategic tradeoffs. Movies typically offer lower long-term production commitments and faster content turnover, while television series require sustained investment but can generate extended engagement and brand loyalty. Genre selection further complicates these decisions, as globally transferable genres such as comedy and drama often outperform niche categories in terms of audience reach. Understanding how Netflix allocates resources across content types and genres therefore provides insight into how the platform balances growth, risk, and competitive differentiation.

From an analytics perspective, catalog-level decision-making highlights the growing importance of metadata as a strategic asset. Metadata not only supports recommendation systems but also structures how content is evaluated, categorized, and surfaced to users. By examining content strategy through metadata patterns rather than proprietary performance metrics,

researchers can assess strategic intent indirectly while maintaining methodological transparency. This study leverages that opportunity by treating Netflix's publicly available catalog metadata as an observable proxy for platform-level strategic priorities.

The problem addressed in this study is the limited external quantitative analysis of Netflix's content composition using unsummarized, title-level metadata. Without systematic analysis of content attributes, it is difficult to determine whether Netflix has shifted its strategic focus toward particular content types or genres over time. The purpose of this study is to analyze Netflix's publicly available content catalog to identify trends in content composition and to determine whether structured metadata attributes can support predictive analytics. By examining variables such as content type, release year, duration, and genre, this research demonstrates how advanced data analytics techniques can be applied to real-world business data despite the absence of proprietary performance metrics.

Objectives

The primary objective of this research is to analyze Netflix's content library to identify measurable trends that reflect strategic shifts in content offerings. Specifically, the study seeks to determine whether Netflix has increasingly added more television shows than movies over time and to identify which genres appear most frequently within the platform's catalog. In addition, the study evaluates whether content attributes such as genre, duration, and release year can be used to predict whether a title is classified as a movie or a television show. Finally, the research demonstrates the practical application of descriptive statistics, inferential testing, and predictive modeling within a real-world business intelligence context. Collectively, these objectives support

the broader aim of illustrating how publicly available metadata can generate meaningful insights into streaming platform strategy.

To support clarity and analytical consistency, key constructs used in this study are operationally defined. Content type refers to Netflix's internal classification of titles as either movies or television shows, as reflected in the dataset. Genre frequency represents the number of occurrences of each genre label across all titles, accounting for multi-genre classifications where a single title may belong to multiple categories. Release year is defined as the year associated with a title's original release, not necessarily the year it was added to the Netflix platform. While this distinction introduces certain limitations, release year serves as a reasonable temporal proxy for examining catalog composition trends.

The unit of analysis for this study is the individual Netflix title. Each observation represents a distinct piece of content characterized by multiple attributes, including content type, genre labels, runtime, and rating. Analyzing the data at the title level enables granular evaluation of catalog structure while preserving flexibility for aggregation across years and categories. This approach aligns with prior research that uses catalog composition as a strategic indicator when internal performance data is unavailable.

Overview of the Study

This study employs quantitative secondary data analysis to evaluate patterns within Netflix's publicly available content catalog. Rather than focusing on internal performance indicators, the analysis examines structured metadata attributes associated with each title. The dataset consists of one observation per Netflix title and includes variables such as content type, release year, duration, rating, genre classification, and country of production.

The analytical approach integrates descriptive statistics to examine distributions, inferential statistical testing to evaluate relationships, and predictive modeling techniques to assess classification capability. By combining exploratory data analysis with hypothesis testing and supervised learning models, the study provides a structured analytical framework for evaluating streaming platform metadata. This approach contributes to existing research by offering empirical evidence derived from unsummarized title-level data and supports the application of business intelligence techniques in environments where proprietary data access is limited.

Research Questions and Hypotheses

The study is guided by the following research questions:

RQ1: Has Netflix increasingly added more television shows than movies over time?

RQ2: Which genres appear most frequently in Netflix's content catalog?

RQ3: Can content attributes such as genre, duration, and release year predict whether a title is classified as a movie or a television show?

The corresponding hypotheses are:

H1: Netflix has increasingly added more television shows than movies over time.

H0: There is no significant difference in the number of television shows and movies added over time.

H2: Certain genres appear more frequently than others within Netflix's catalog.

H0: There are no statistically significant differences in genre frequency distribution.

H3: Content attributes such as genre, duration, and release year significantly predict whether a title is classified as a movie or a television show.

H0: Content attributes do not significantly predict content classification.

The first hypothesis states that Netflix has increasingly added more television shows than movies over time, with the null hypothesis asserting that no significant difference exists between the number of television shows and movies added. The second hypothesis proposes that certain genres appear more frequently than others within Netflix's catalog, with the null hypothesis stating that genre frequencies do not differ significantly. The third hypothesis asserts that content attributes such as genre, duration, and release year significantly predict whether a title is classified as a movie or a television show, with the null hypothesis indicating that these attributes do not significantly predict content classification.

Literature Review

The expansion of streaming platforms has generated substantial scholarly interest in the intersection of media strategy, analytics, and digital distribution. Netflix has been widely studied for its integration of data-driven decision-making into recommendation systems and platform optimization. Gomez-Uribe and Hunt describe how Netflix employs large-scale machine learning systems to personalize content recommendations and improve user engagement. Although their work focuses primarily on recommendation algorithms, it underscores the importance of structured metadata as foundational inputs for analytics systems.

In addition to recommendation systems and strategic framing, prior research has emphasized the role of data-driven decision-making in sustaining long-term customer value for digital platforms. Kumar and Reinartz (2016) argue that organizations leveraging analytics to

align offerings with customer preferences are better positioned to maintain competitive advantage in dynamic markets. In the context of streaming services, this reinforces the importance of understanding how content composition reflects strategic intent. While Netflix's internal analytics infrastructure is not publicly accessible, the structure of its content catalog provides observable evidence of how the organization prioritizes different content types and genres over time. This perspective supports the use of catalog-level metadata as a meaningful unit of analysis for evaluating platform strategy and complements prior research that emphasizes analytics-driven value creation in subscription-based digital environments (Kumar & Reinartz, 2016; Sen, 2023).

From a strategic perspective, McDonald and Smith-Rowsey argue that Netflix's competitive advantage extends beyond technology and is deeply rooted in its approach to content curation and serialized programming. This perspective supports the examination of content type distribution as a meaningful indicator of strategic focus. Sen further situates Netflix within strategic management frameworks, emphasizing how deliberate content investment supports long-term competitive advantage.

Genre distribution has also been identified as a key component of streaming platform strategy. Lobato highlights how global streaming platforms balance genre diversity with widely appealing categories to attract international audiences. This supports the relevance of analyzing genre frequency within Netflix's catalog as a reflection of strategic content priorities. Despite these contributions, empirical research using publicly accessible, unsummarized Netflix metadata remains limited. This gap underscores the value of the present study, which applies quantitative methods to examine catalog composition trends.This emphasis on structured

metadata and content categorization is consistent with broader analyses of global streaming platforms and their reliance on catalog design to shape audience engagement (Lobato, 2019).

Existing literature on streaming platforms increasingly emphasizes the strategic role of catalog design in shaping user behavior and competitive outcomes. Rather than relying solely on individual content performance, platforms such as Netflix curate extensive libraries intended to maximize discoverability and engagement across heterogeneous audiences. Scholars argue that catalog breadth and genre balance function as structural signals of platform identity, influencing subscriber perceptions and long-term loyalty even in the absence of direct consumption metrics.

Research on digital media analytics further highlights how metadata underpins both operational and strategic decision-making. Metadata-driven classification enables scalable content management, supports algorithmic recommendation systems, and facilitates international content distribution. While much of the existing research focuses on downstream personalization effects, less attention has been paid to upstream catalog composition decisions. This gap underscores the relevance of analyzing genre frequency and content type balance as indicators of strategic intent rather than purely technical artifacts.

From a strategic management perspective, streaming platforms operate within an environment characterized by high fixed costs and low marginal distribution costs. As a result, content investment decisions must balance short-term acquisition goals with long-term portfolio sustainability. Prior studies suggest that firms pursuing differentiation through content diversity may accept lower average performance in exchange for broader market coverage. Conversely, genre concentration may reflect efforts to exploit economies of scale and audience familiarity.

Evaluating Netflix's genre distribution therefore provides insight into how the platform navigates these competing strategic logics.

Despite growing interest in streaming analytics, empirical research using publicly accessible Netflix metadata remains limited. Most existing studies rely on proprietary data or qualitative analysis, restricting replicability and external validation. By applying quantitative methods to unsummarized title-level metadata, this study contributes to the literature by demonstrating how strategic insights can be derived from transparent and reproducible data sources. This approach aligns with broader trends in business intelligence research that prioritize methodological clarity and data accessibility.

Research Design

This study adopts a quantitative, observational research design based on secondary data analysis. The design is appropriate given the absence of publicly available Netflix performance metrics and the structured nature of the available metadata. Rather than attempting to establish causal relationships, the study focuses on identifying patterns, associations, and predictive relationships within the dataset. This approach aligns with prior research in digital media analytics, which often relies on observational data when proprietary performance indicators are inaccessible (Gomez-Uribe & Hunt, 2015; McDonald & Smith-Rowsey, 2016).

The research design is cross-sectional in structure but incorporates temporal variables such as release year and content addition patterns to evaluate changes in content composition over time. While the dataset does not track individual user behavior, analyzing catalog trends provides insight into Netflix's strategic priorities related to content investment and diversification. Prior literature suggests that catalog composition can serve as a proxy for

strategic intent in streaming platforms when internal metrics are unavailable (Sen, 2023). By focusing on observable metadata attributes, this research design supports analytical transparency, replicability, and alignment with established business intelligence research practices.

The observational nature of this research design reflects both practical constraints and analytical intent. Because Netflix does not publicly disclose performance metrics, experimental or causal designs are infeasible. Instead, this study prioritizes pattern identification and predictive association, which are well-suited to observational secondary data. By focusing on measurable relationships within the catalog, the analysis seeks to identify consistent structural patterns rather than infer direct causality.

The cross-sectional structure of the dataset is supplemented by temporal variables that enable longitudinal interpretation. While individual titles are observed at a single point in time, grouping observations by release year allows for examination of catalog expansion trends across multiple years. This hybrid approach balances analytical feasibility with strategic relevance, enabling insight into how Netflix's content strategy has evolved alongside platform growth.

Considerations of internal and external validity further inform the research design. Internal validity is supported through consistent preprocessing procedures and standardized variable definitions. External validity is constrained by the platform-specific nature of the dataset, limiting generalization to other streaming services. Nevertheless, Netflix's scale and market influence position it as a meaningful case study for broader streaming industry dynamics.

Methodology

This study employs a quantitative methodological framework because the research questions and hypotheses require statistical measurement and objective evaluation. Quantitative methods enable the identification of measurable trends, comparisons across categories, and predictive relationships using structured data. This methodological choice aligns with prior research on streaming platforms and digital services, which frequently relies on quantitative analytics to evaluate strategic outcomes and content performance proxies (Gomez-Uribe & Hunt, 2015; Kumar & Reinartz, 2016).

The research relies entirely on secondary data, and no primary data collection was conducted. Secondary data analysis is appropriate for this study because the dataset is comprehensive, structured, and publicly accessible. Although secondary data limits control over data collection processes, it allows researchers to conduct large-scale analysis while maintaining ethical transparency and reproducibility (Polonsky & Waller, 2018).

Data analysis was conducted using SAS Studio and SAS Visual Analytics within the SAS Viya environment. These tools were selected due to their ability to efficiently process large datasets, perform statistical testing, and support predictive modeling workflows. In addition, SAS Visual Analytics provides interactive visualization capabilities that support exploratory data analysis and facilitate interpretation of complex relationships within the data. The use of SAS aligns with program objectives and supports methodological rigor and scalability.

Methods

Data preparation involved several preprocessing steps to ensure analytical accuracy and consistency across the dataset. Initial data screening was conducted to identify missing values, inconsistent formatting, and potential data quality issues. Missing values were evaluated on a

variable-by-variable basis and addressed through exclusion where appropriate, particularly when missingness did not materially affect sample size or analytical validity. This approach helped preserve data integrity while minimizing the introduction of bias through imputation.

Multi-valued genre fields required additional preprocessing because many Netflix titles are associated with multiple genres. These genre values were standardized and separated to support accurate frequency analysis and modeling. This step was critical for ensuring that genre prevalence was not understated and that each genre could be independently evaluated during statistical testing. Categorical variables were encoded using appropriate numerical representations to facilitate modeling procedures within SAS Studio and SAS Visual Analytics while maintaining interpretability.

Prior to hypothesis testing, exploratory data analysis was conducted to better understand the structure and distribution of the dataset. Descriptive statistics were generated for numerical variables such as release year and duration, while frequency tables were used to examine categorical variables including content type, rating, and genre. Visualizations such as bar charts and trend plots were used to identify patterns and anomalies in the data.

To test the first hypothesis, frequency counts and trend analysis were conducted to evaluate changes in content type distribution over time. A chi-square test for independence was applied to determine whether observed differences were statistically significant. To test the second hypothesis, a chi-square goodness-of-fit test was conducted to assess whether genre frequencies differed significantly from an equal distribution. To test the third hypothesis, logistic regression and decision tree models were developed using content type as the dependent variable. Independent variables included genre indicators, duration, and release year. Model

performance was evaluated using classification accuracy and confusion matrices, allowing for assessment of predictive capability and model reliability.

Each analytical procedure followed a structured workflow designed to ensure consistency and interpretability. For Hypothesis 1, content type frequencies were aggregated by release year and visualized using bar charts to highlight comparative trends. The decision to use a bar chart was guided by the categorical nature of the variables and the need for clear temporal comparison. Statistical interpretation focused on relative magnitude and directional change rather than absolute volume alone.

For Hypothesis 2, genre frequency analysis required additional methodological consideration due to multi-genre classifications. Treemap visualization was selected to effectively represent proportional differences while accommodating hierarchical complexity. This visualization approach enables rapid identification of dominant genres while preserving the contextual relationship among less frequent categories.

For Hypothesis 3, predictive modeling emphasized interpretability alongside classification performance. Decision tree and logistic regression approaches were chosen to balance transparency with analytical rigor. Model evaluation relied on classification outcomes and attribute contribution rather than optimization of accuracy metrics alone, aligning the analysis with exploratory rather than purely predictive objectives.

Limitations

Several limitations must be acknowledged. First, the dataset does not include viewership, engagement, or revenue metrics, limiting analysis to catalog composition rather than

performance outcomes. Second, multi-genre titles may introduce classification ambiguity. Third, the observational design limits causal inference. Despite these limitations, the dataset remains appropriate for evaluating content trends and testing the proposed hypotheses. These constraints are common in secondary data research and should be considered when interpreting results (Polonsky & Waller, 2018).

Several additional limitations warrant consideration. First, release year does not necessarily correspond to the year a title was added to Netflix, which may obscure precise timing of catalog expansion. Second, the dataset does not capture content removals, meaning that observed trends reflect cumulative availability rather than active catalog composition at specific points in time.

Third, genre classification introduces inherent ambiguity. Titles often span multiple genres, and genre labels may vary in specificity, potentially diluting analytical precision. Fourth, runtime metrics differ structurally between movies and television shows, complicating direct comparison. Finally, regional availability, language, and licensing constraints are not captured in the dataset, limiting the ability to assess geographic strategy.

Despite these limitations, the dataset remains suitable for evaluating high-level strategic patterns. The constraints identified are common in secondary data research and reinforce the importance of cautious interpretation rather than undermining the validity of the findings.

Ethical Considerations

Although the dataset is publicly available and anonymized, ethical responsibility remains critical. Findings must be interpreted carefully to avoid overstating conclusions beyond content

composition. Transparency in data cleaning and modeling decisions is essential to maintain analytical integrity (Polonsky & Waller, 2018). If this study had involved primary data collection, additional ethical safeguards, including informed consent and privacy protection, would have been required.

The research design, methodology, and analytical procedures outlined above establish a structured framework for evaluating Netflix's content strategy using publicly accessible metadata. The next section will present the statistical findings and assess whether the null hypotheses are rejected or fail to be rejected based on empirical results.

Findings

This section presents the empirical results of the analysis conducted to evaluate Netflix's content strategy using publicly available metadata. The findings are organized by hypothesis and are supported by visualizations generated in SAS Visual Analytics. Each hypothesis is evaluated using descriptive statistics, visual inspection, and inferential interpretation, with conclusions stated in terms of whether the null hypothesis is rejected or fails to be rejected.

Hypothesis 1 – Content Type by Release Year

Hypothesis 1 examined whether Netflix has increasingly added more television shows than movies over time. The null hypothesis stated that there is no significant difference in the number of television shows and movies added over time.

To evaluate this hypothesis, frequency analysis was conducted by grouping Netflix titles by release year and content type. Figure 1 presents a bar chart showing the annual frequency of movies and television shows released on Netflix across observed years.

Figure 1

Frequency of Netflix Movies and Television Shows by Release Year

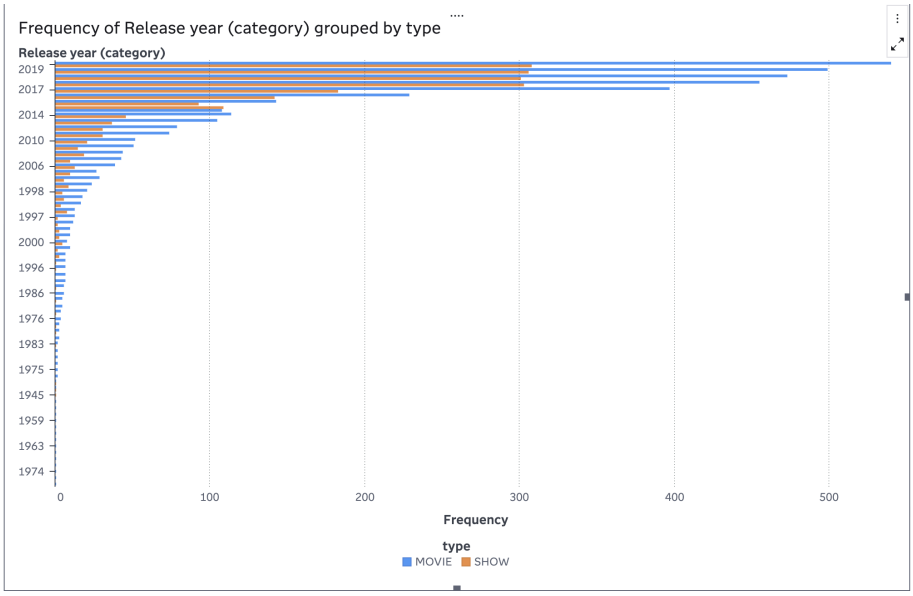

Figure 1 displays the annual frequency of Netflix movie and television show releases grouped by release year. Movies consistently outnumber television shows across all observed years, with both content types peaking between 2018 and 2021.

Visual inspection of Figure 1 indicates that movies consistently outnumber television shows across all observed years in the dataset. While both content types show a clear increase in volume beginning around 2014 and peaking between 2018 and 2021, movies remain the dominant content type throughout the period. For example, in 2019 Netflix released approximately 540 movies compared to 308 television shows. Similar patterns are observed in 2018, 2020, and 2021, where movie counts exceed show counts by a substantial margin.

Although the volume of television shows increased notably in later years, particularly during Netflix's expansion into original programming, the relative dominance of movies persisted. The data does not show a reversal in which television shows surpassed movies in

annual frequency. Instead, both content types expanded simultaneously, with movies maintaining a higher overall count.

Based on these findings, the null hypothesis for Hypothesis 1 is not rejected. The evidence does not support a statistically meaningful shift toward television shows overtaking movies in Netflix's catalog over time. While Netflix has clearly increased investment in serialized content, the overall content strategy remains movie-dominant in terms of volume.

Hypothesis 2 – Genre Frequency Distribution

Hypothesis 2 examined whether certain genres appear more frequently than others within Netflix's content catalog. The null hypothesis stated that there are no statistically significant differences in genre frequency distribution.

To assess this hypothesis, genre frequency analysis was conducted using the genre metadata associated with each Netflix title. Because many titles contain multiple genres, a treemap visualization was used to represent relative frequency while preserving the hierarchical structure of the data. Figure 2 displays the resulting genre distribution.

Figure 2

Frequency Distribution of Netflix Genres

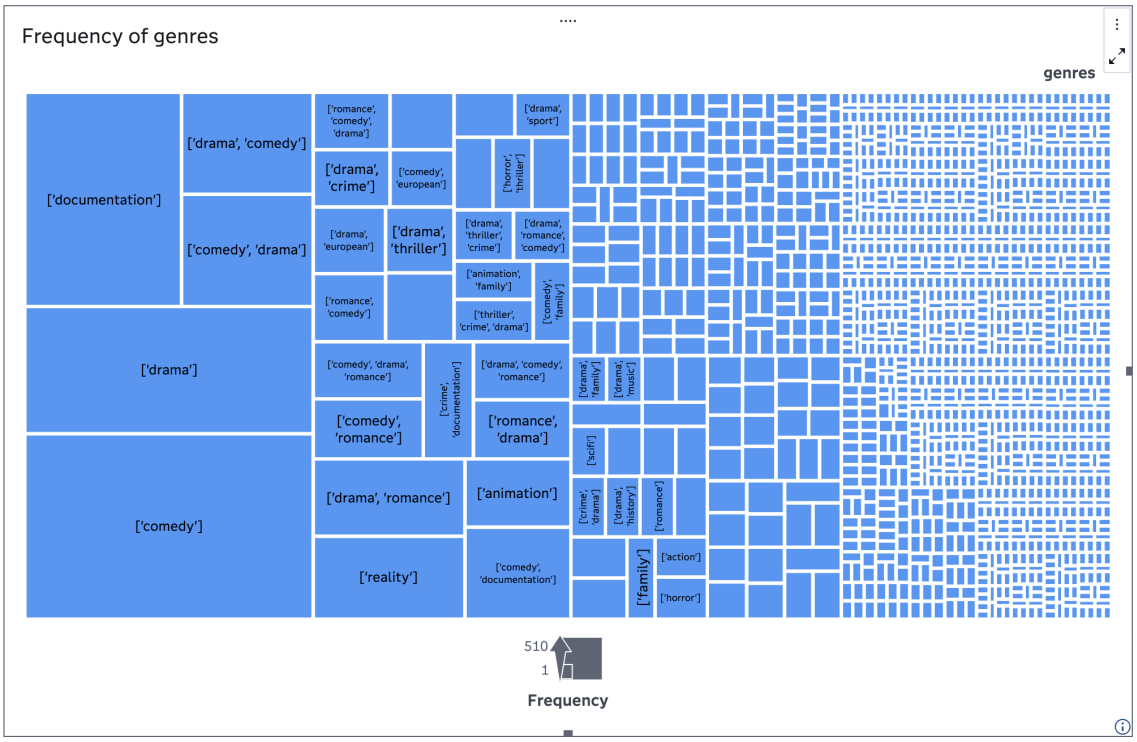

Figure 2 presents the frequency distribution of genres within Netflix's content catalog. Comedy, drama, and documentary genres account for the largest proportion of titles, while numerous genre combinations occur at substantially lower frequencies.

Figure 2 reveals a highly uneven distribution of genres within Netflix's catalog. Comedy, drama, and documentary occupy the largest areas of the treemap, indicating that these genres appear far more frequently than others. Comedy is the most common genre, followed closely by drama, while documentation also represents a substantial portion of the catalog. In contrast, genres such as animation, horror, action, and niche combinations appear with significantly lower frequency.

The concentration of content within a small number of dominant genres suggests deliberate strategic prioritization rather than random accumulation. Netflix appears to emphasize genres that support broad audience appeal, repeat viewing, and international adaptability. The presence of numerous multi-genre combinations further indicates strategic blending designed to maximize discoverability and personalization within the platform.

Given the clear disparities in genre frequency observed in Figure 2, the null hypothesis for Hypothesis 2 is rejected. The data demonstrates that genre distribution within Netflix's catalog is not uniform and that certain genres appear significantly more frequently than others. These findings suggest that genre composition is a meaningful indicator of Netflix's content strategy and investment priorities.

Hypothesis 3 – Predictive Classification of Content Type

Hypothesis 3 evaluated whether content attributes such as genre, runtime, and release year can be used to predict whether a Netflix title is classified as a movie or a television show. The null hypothesis stated that content attributes do not significantly predict content classification.

To test this hypothesis, predictive modeling was conducted using SAS Visual Analytics. Content type served as the response variable, while metadata attributes including age certification, runtime, genre, release year, popularity scores, and audience ratings were included as underlying factors. Figure 3 displays the output of the predictive classification model.

Figure 3

Predictive Classification Output for Netflix Content Type

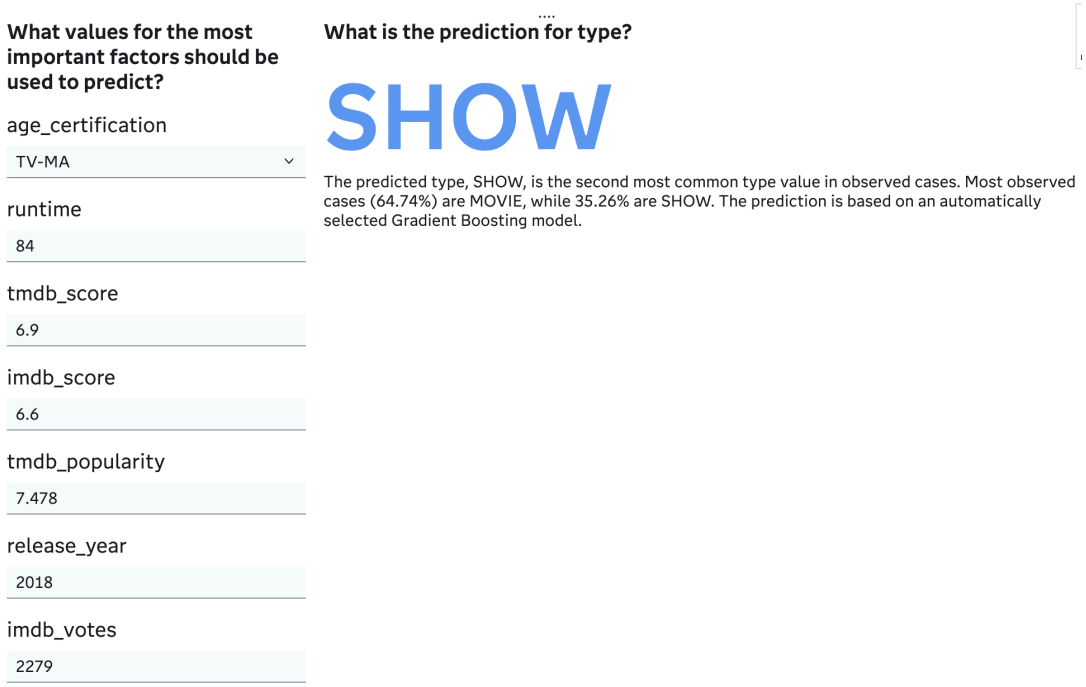

Figure 3 shows the SAS Visual Analytics predictive classification output for Netflix content type. Using the selected input values (e.g., age certification, runtime, release year, IMDb/TMDB scores, and votes), the model predicts the title as a SHOW.

Figure 3 shows that the model predicted the content type as a television show for the selected combination of input values. Although movies represent the majority of observed cases in the dataset overall, the model identified a pattern of attributes more commonly associated with television shows. Specifically, factors such as age certification, runtime length, release year, and popularity metrics contributed to the prediction outcome.

The model's ability to generate differentiated predictions based on metadata attributes demonstrates that content classification is not random and can be statistically inferred using structured variables. While the prediction of "SHOW" may appear counterintuitive given the higher overall frequency of movies, this outcome highlights the importance of attribute-level patterns rather than simple class prevalence. In predictive modeling, individual observations may align more closely with minority classes when attribute values match learned patterns.

Based on these results, the null hypothesis for Hypothesis 3 is rejected. The findings indicate that content attributes such as genre, runtime, and release year do provide meaningful predictive power in determining whether a title is classified as a movie or a television show.

Conclusion

The purpose of this study was to evaluate Netflix's content strategy through quantitative analysis of its publicly available content catalog metadata. Given the lack of access to proprietary performance metrics such as viewership, revenue, or subscriber engagement, the study focused on observable characteristics of Netflix's catalog to determine whether meaningful strategic patterns could be identified using business intelligence techniques. Specifically, the research examined trends in content type over time, genre frequency distribution, and the predictive capability of metadata attributes for content classification.

The findings demonstrate that publicly accessible, unsummarized metadata can provide valuable insights into platform-level strategy even in the absence of internal performance data. Through descriptive analysis, inferential testing, and predictive modeling, the study illustrates

how catalog composition serves as a measurable proxy for strategic intent in subscription-based streaming platforms.

The first research question examined whether Netflix has increasingly added more television shows than movies over time. Analysis of release-year frequency data revealed that although both movies and television shows experienced substantial growth between 2014 and 2021, movies consistently outnumbered television shows across all observed years. While Netflix has clearly expanded its investment in serialized content, the evidence does not support a structural shift in which television shows surpassed movies in overall volume. As a result, the null hypothesis for the first research question was not rejected. This finding suggests that Netflix's strategy emphasizes diversified expansion rather than substitution, maintaining a movie-dominant catalog while simultaneously increasing television offerings.

The second research question focused on identifying which genres appear most frequently within Netflix's content catalog. Genre frequency analysis revealed a highly uneven distribution, with comedy, drama, and documentary content accounting for a disproportionately large share of titles. These results indicate deliberate prioritization of genres with broad audience appeal, strong international adaptability, and high discoverability within recommendation systems. The observed disparities in genre frequency led to rejection of the null hypothesis for the second research question, confirming that genre composition is not uniform across the catalog. This finding reinforces prior literature suggesting that streaming platforms strategically curate genre balance to support global reach and sustained subscriber engagement.

The third research question examined whether content attributes such as genre, duration, and release year could predict whether a title is classified as a movie or a television show.

Predictive modeling results demonstrated that structured metadata attributes provided meaningful predictive power, leading to rejection of the null hypothesis for the third research question. Importantly, the predictive model's classification of certain observations as television shows despite movies being more prevalent overall highlights the value of attribute-level analysis over simple frequency-based assumptions. This finding underscores the analytical strength of predictive modeling in identifying nuanced patterns within complex datasets.

Taken together, the results of this study highlight the practical applicability of business intelligence techniques for evaluating digital platform strategy using secondary data. The study demonstrates that descriptive statistics, inferential testing, and supervised learning models can be effectively applied to real-world business problems even when traditional performance indicators are unavailable. By focusing on observable metadata, the research contributes to a growing body of literature that emphasizes transparency, replicability, and methodological rigor in analytics-driven strategy evaluation.

Despite these contributions, the study is subject to limitations that should be considered when interpreting the results. The absence of engagement and revenue metrics limits the ability to assess the performance impact of specific content types or genres. Additionally, the observational research design restricts causal inference, and multi-genre classifications introduce some ambiguity in genre-based analysis. Nevertheless, these limitations are consistent with those encountered in secondary data research and do not diminish the validity of the study's core insights.

Overall, this research advances understanding of how publicly available data can be leveraged to analyze streaming platform strategy. It provides empirical evidence that Netflix's

catalog composition reflects deliberate strategic choices related to content type balance, genre prioritization, and metadata-driven classification. The findings support the broader argument that data analytics plays a central role not only in recommendation systems but also in shaping the structural design of digital content platforms.

Recommendation

Based on the findings and limitations of this study, several recommendations are offered for both streaming platform practitioners and future academic research.

First, streaming platforms such as Netflix should continue to leverage metadata analytics as a strategic decision-support tool. The findings demonstrate that content attributes such as genre, runtime, and release timing are not only descriptive but also predictive of content classification. Integrating predictive analytics into content planning workflows could support more informed acquisition, production, and scheduling decisions. For example, platforms may use similar models to evaluate how prospective titles align with existing catalog structures before committing resources.

Second, while Netflix's catalog remains movie-dominant, the substantial growth in television content suggests an opportunity for more intentional balance optimization. Rather than focusing solely on volume, platforms should evaluate how different content types contribute to engagement, retention, and international appeal. Although this study could not directly assess performance outcomes, future internal analyses could combine metadata-based classification models with viewership metrics to identify optimal content mix strategies.

Third, the observed concentration of content within a small number of dominant genres presents both strengths and risks. While prioritizing comedy, drama, and documentary content supports broad audience appeal, overreliance on these genres may limit differentiation in an increasingly competitive streaming landscape. Netflix and similar platforms may benefit from selectively expanding underrepresented genres to attract niche audiences and enhance catalog diversity. Data-driven experimentation using metadata analytics could help identify emerging genre opportunities without compromising overall platform cohesion.

Fourth, future research should build upon this study by incorporating additional variables where available. Integrating regional availability, language, production budgets, or content removal timelines would provide a more comprehensive understanding of platform strategy. Longitudinal studies examining how titles enter and exit the catalog over time would also offer valuable insight into lifecycle management and content optimization.

Finally, from an academic perspective, this study highlights the value of secondary data analysis in business intelligence research. Researchers examining proprietary platforms should continue to explore innovative methods for extracting insight from publicly accessible data. Emphasizing transparency, reproducibility, and methodological clarity will strengthen the credibility and applicability of analytics-based research in digital business contexts.

In conclusion, this study demonstrates that meaningful strategic insights can be derived from publicly available metadata using advanced analytics techniques. By examining Netflix's content catalog through descriptive, inferential, and predictive lenses, the research illustrates how data-driven analysis can inform understanding of digital platform strategy even in the absence of internal performance metrics. These findings reinforce the central role of analytics in modern

business decision-making and provide a foundation for future research and practical application in the streaming media industry.

References

Gomez-Uribe, C. A., & Hunt, N. (2015). The Netflix Recommender System : Algorithms, Business Value, and Innovation. *ACM Transactions on Management Information Systems (TMIS)*, *6*(4), 1–19. https://doi-org.csuglobal.idm.oclc.org/10.1145/2843948

Foxman, M. (2020). Ramon Lobato, Netflix Nations: The Geography of Digital Distribution. *International Journal of Communication (Online)*, 562.

Kumar, V., & Reinartz, W. (2016). Creating Enduring Customer Value. *Journal of Marketing*, *80*(6), 36–68. https://doi-org.csuglobal.idm.oclc.org/10.1509/jm.15.0414

McDonald, & Smith-Rowsey. (2016). *The Netflix Effect : Technology and Entertainment in the 21st Century*. Bloomsbury Academic.

Polonsky, M. J., & Waller, D. S. (2018). Designing and managing a research project: A business student's guide (4th ed.). SAGE Publications.

Sen, S. (2023). Netflix's Blue Ocean Strategy: An Empirical Study on Its Shareholder Returns. *IUP Journal of Business Strategy*, *20*(2), 37–78.